



SUMMARY

This circuit is designed to monitor soil moisture levels and control a water pump based on the moisture readings. It also includes environmental sensors for light and atmospheric conditions, and it can communicate data via an LTE module. The circuit is powered by a 12V battery, regulated to 5V for most components. An Arduino UNO serves as the central microcontroller, interfacing with various sensors and actuators.



COMPONENT LIST

1. Arduino UNO

- **Description:** A microcontroller board based on the ATmega328P.
- **Pins:** 30 digital input/output pins, 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header, and a reset button.

2. Soil Moisture V2

- **Description:** A sensor used to measure the volumetric content of water in soil.
- **Pins:** GND, VCC, OUT

3. a7670E LTE Module

- **Description:** A module for LTE communication.
- **Pins:** GROUND, RX, TX, K, VCC, S

4. 5V Relay

- **Description:** An electrically operated switch used to control a high-power device with a low-power signal.
- **Pins:** NOpen, Common, NClosed, VCC, -, S

5. Resistors

- **Description:** Passive electrical components that implement electrical resistance as a circuit element.
 - Resistor 1: 330 Ohms
 - Resistor 2: 1000 Ohms
 - Resistor 3: 2000 Ohms

6. Regulator

- **Description:** A device that maintains a constant voltage level.
- **Pins:** Vin+, Vin-, Vout+, Vout-

7. Electrolytic Capacitor

- **Description:** A type of capacitor that uses an electrolyte to achieve a larger capacitance.
- **Capacitance:** 0.00047 Farads
- **Pins:** -, +

8. Ceramic Capacitor

- **Description:** A capacitor with a ceramic dielectric.
- **Capacitance:** 0.0000001 Farads
- **Pins:** pin0, pin1

9. BH1750

- **Description:** A digital light sensor.
- **Pins:** ADDR, SDA, SCL, GND, VCC



COMPONENT LIST

10. BME280

- **Description:** A sensor for measuring atmospheric pressure and temperature.
- **Pins:** VIN, GND, SCL, SDA

11. 4G Antenna

- **Description:** An antenna for LTE communication.
- **Pins:** Connection

12. Red LED

- **Description:** A light-emitting diode that emits red light.
- **Pins:** +, -

13. 12V Battery (mini)

- **Description:** A power source for the circuit.
- **Pins:** -, +

14. Water Pump

- **Description:** A device used to move water.
- **Pins:** VCC, GND



WIRING DETAILS



Arduino UNO

- **GND:** Connected to the common ground net.
- **5V:** Connected to the VCC net for powering components.
- **A0:** Connected to the OUT pin of the Soil Moisture V2.
- **A4:** Connected to the SDA pins of BH1750 and BME BMP280.
- **A5:** Connected to the SCL pins of BH1750 and BME BMP280.
- **D7:** Connected to the S pin of the 5V Relay.
- **D10:** Connected to the TX pin of the a7670E LTE Module.
- **D11:** Connected to pin1 of the 1000 Ohm Resistor.



Soil Moisture V2

- **GND:** Connected to the common ground net.
- **VCC:** Connected to the VCC net.
- **OUT:** Connected to A0 on the Arduino UNO.



a7670E LTE Module

- **GROUND:** Connected to the common ground net.
- **RX:** Connected to pin2 of the 2000 Ohm Resistor.
- **TX:** Connected to D10 on the Arduino UNO.
- **VCC:** Connected to the VCC net.
- **S:** Connected to the Connection pin of the 4G Antenna.



5V Relay



WIRING DETAILS



1. 5V Relay

- **NOpen:** Connected to pin1 of the 330 Ohm Resistor.
- **Common:** Connected to the VCC net.
- **VCC:** Connected to the VCC net.
- **-:** Connected to the common ground net.
- **S:** Connected to D7 on the Arduino UNO.



2. Resistors

- **330 Ohm Resistor:** pin1 connected to NOpen of the 5V Relay, pin2 connected to the + pin of the Red LED.
- **1000 Ohm Resistor:** pin1 connected to D11 on the Arduino UNO, pin2 connected to pin2 of the 2000 Ohm Resistor.
- **2000 Ohm Resistor:** pin1 connected to the common ground net, pin2 connected to RX of the a7670E LTE Module.



3. Regulator

- **Vin+:** Connected to the + pin of the 12V Battery.
- **Vin-:** Connected to the - pin of the 12V Battery.
- **Vout+:** Connected to the VCC net.
- **Vout-:** Connected to the common ground net.



4. Electrolytic Capacitor

- **-:** Connected to the common ground net.
- **+:** Connected to the VCC net.



5. Ceramic Capacitor

- **pin0:** Connected to the VCC net.
- **pin1:** Connected to the common ground net.



6. BH1750

- **GND:** Connected to the common ground net.
- **VCC:** Connected to the VCC net.
- **SDA:** Connected to A4 on the Arduino UNO.
- **SCL:** Connected to A5 on the Arduino UNO.



7. BME280

- **GND:** Connected to the common ground net.
- **VIN:** Connected to the VCC net.
- **SDA:** Connected to A4 on the Arduino UNO.
- **SCL:** Connected to A5 on the Arduino UNO.



8. 4G Antenna



WIRING DETAILS

1. 4G Antenna

- **Connection:** Connected to the S pin of the a7670E LTE Module.

2. Red LED

- **+**: Connected to pin2 of the 330 Ohm Resistor.
- **-**: Connected to the common ground net.

3. 12V Battery (mini)

- **+**: Connected to Vin+ of the Regulator.
- **-**: Connected to Vin- of the Regulator.

4. Water Pump

- **VCC:** Connected to NOpen of the 5V Relay.
- **GND:** Connected to the common ground net.